0.1 Connectedness, Path-Connectedness, and the Preservation of Topological Properties

Unlike compactness, which relates to bounding and covering, connectedness describes the structural indivisibility of a space. We define this property negatively, via the inability to separate the space into disjoint open components.

A metric space (X, d) is **disconnected** if there exist two non-empty, disjoint open sets $U, V \subset X$ such that $X = U \cup V$. The space (X, d) is **connected** if it is not disconnected. A subset $E \subset X$ is connected if it is connected as a metric subspace under the inherited metric.

A more intuitive, kinesthetic notion of connectedness relies on the existence of continuous paths between any two points.

A metric space (X, d) is **path-connected** if for every pair of points $x, y \in X$, there exists a continuous function $\gamma : [0, 1] \to X$ such that $\gamma(0) = x$ and $\gamma(1) = y$.

Every path-connected metric space (X, d) is connected.

We proceed by contradiction. Assume (X, d) is path-connected but disconnected. By definition, there exist non-empty, disjoint open sets U and V such that $X = U \cup V$. Because U and V are non-empty, we can choose points $x \in U$ and $y \in V$.

Since X is path-connected, there exists a continuous path $\gamma : [0, 1] \to X$ such that $\gamma(0) = x$ and $\gamma(1) = y$. Consider the preimages of U and V under γ:

$$A = \gamma^{-1}(U) = \{t \in [0, 1] \mid \gamma(t) \in U\}$$

$$B = \gamma^{-1}(V) = \{t \in [0, 1] \mid \gamma(t) \in V\}$$

Because γ is a continuous mapping, the preimages of open sets are open in the relative topology of $[0, 1]$. Thus, A and B are open subsets of $[0, 1]$. Furthermore, $0 \in A$ (since $\gamma(0) = x \in U$) and $1 \in B$ (since $\gamma(1) = y \in V$), making both sets non-empty. Since $U \cap V = \emptyset$, it strictly follows that $A \cap B = \emptyset$. Finally, because $X = U \cup V$, every $t \in [0, 1]$ maps into either U or V, meaning $A \cup B = [0, 1]$.

This implies that $[0, 1]$ is the union of two non-empty, disjoint open sets A and B, which means $[0, 1]$ is disconnected. However, the interval $[0, 1] \subset R$ is fundamentally connected (a direct consequence of the Completeness Axiom established in Chapter 1). This contradiction forces us to conclude that X must be connected.

The power of topological invariants lies in their preservation under continuous mappings. Connectedness is perfectly preserved, which immediately yields the generalized Intermediate Value Theorem.

[Preservation of Connectedness] Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \to Y$ be a continuous function. If X is connected, then the image $f(X)$ is connected in Y.

Suppose, for the sake of contradiction, that $f(X)$ is disconnected. Then there exist open sets U_Y and V_Y in Y such that $U = U_Y \cap f(X)$ and $V = V_Y \cap f(X)$ are non-empty, disjoint open sets in the subspace topology of $f(X)$, and $f(X) = U \cup V$.

Consider the preimages in X:

$$A = f^{-1}(U) \quad and \quad B = f^{-1}(V)$$

Because f is continuous, A and B are open in X. Since U and V are non-empty and contained in the image of f, both A and B are non-empty. Because $U \cap V = \emptyset$, a single point in X cannot map to both U and V, rendering $A \cap B = \emptyset$. Lastly, because $f(X) = U \cup V$, every point in X maps to either U or V, so $X = A \cup B$. This constitutes a disconnection of X, contradicting the hypothesis that X is connected. Thus, $f(X)$ must be connected.

[Generalized Intermediate Value Theorem] Let (X, d) be a connected metric space, and let $f : X \to R$ be continuous. If $a, b \in X$ such that $f(a) < f(b)$, then for every real number c satisfying $f(a) < c < f(b)$, there exists a point $x \in X$ such that $f(x) = c$.

By Theorem 4.3.2, the image $f(X) \subset R$ is connected. The only connected subsets of the real line are intervals (rays, segments, or R itself). Since $f(a) \in f(X)$ and $f(b) \in f(X)$, the entire segment $[f(a), f(b)]$ must be contained within the interval $f(X)$. Because $c \in [f(a), f(b)]$, it follows that $c \in f(X)$. By the definition of the image, there exists at least one $x \in X$ such that $f(x) = c$.